Interpretable Federated Kolmogorov–Arnold Surrogates for Autonomous and Evolutive Resource Optimization over Non-Stationary LEO Links

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***Abstract*—Low Earth orbit (LEO) networks must repeatedly solve resource-allocation problems whose optima have no closed form and that drift as satellites pass and interference regimes switch. We study an autonomous and evolutive approach in which a federated Kolmogorov–Arnold network (KAN) learns the *solution map* of an energy-efficiency transmit-power optimization, mapping a link state to its optimal power online. Our claim is not a raw performance win: because the energy-efficiency objective is flat near its optimum, the achieved utility regret is comparable across surrogates. Instead, we show that the KAN is the more *trustworthy* surrogate for autonomous, non-stationary, bandwidth-limited deployment along four axes an MLP does not jointly provide: (i) it is more sample-efficient on the solution map (about $1.8\times$ lower normalized error at equal parameters, Wilcoxon $p < 0.01$, while a linear predictor fails entirely); (ii) it extrapolates to unseen interference regimes about an order of magnitude more accurately; (iii) its additive-spline structure yields closed-form, auditable design rules; and (iv) a contextual-bandit controller that learns the per-slot update sparsity under the communication budget halves the uplink at competitive accuracy. We also report transparent negatives: online grid self-evolution gives no significant gain over a static grid, and the accuracy advantage does not survive large channel-estimation error. The study positions interpretable KAN surrogates as a practical building block for autonomous optimization in non-stationary networked-AI systems.**

***Index Terms*—Kolmogorov–Arnold networks, federated learning, learning to optimize, LEO satellite networks, energy efficiency, interpretability, non-stationary environments.**

I. Introduction

NEXT-generation non-terrestrial networks built on LEO constellations operate under fast, non-stationary conditions: the channel gain sweeps with elevation across a pass, and handovers between satellites and beams switch the interference regime on a timescale of seconds [1], [2]. Each terminal must continually re-solve resource-allocation problems such as energy-efficient power control, whose optimizers have no closed form and must be recomputed as conditions drift. Solving these from scratch online is costly, and the time-varying feasible region defeats precomputed lookups [3].

This setting raises three coupled challenges. First, the per-instance optimum has no closed form, so the system needs a fast yet accurate solver. Second, terminals are numerous and bandwidth to the satellite is scarce, so any learned solver must be trained in a federated, communication-frugal manner. Third, the environment is non-stationary: as the geometry and interference drift, a model fit once becomes stale, and operators increasingly require the deployed policy to be auditable rather than a black box.

A learning-to-optimize view replaces the per-instance solver with a surrogate that maps a problem state directly to its optimal decision [3]. For this to be trustworthy in a networked-AI system it should be (i) accurate on the deterministic solution map, (ii) compact and communication-light when trained in a federated manner across many terminals [4], [5], and (iii) interpretable, so operators can audit the learned policy. A plain multilayer perceptron (MLP) meets the first two only partially and fails the third outright.

We argue that Kolmogorov–Arnold networks (KANs) [6] are a natural fit. A KAN places learnable univariate spline functions on its edges, so a trained model is a sum of one-dimensional curves that can be inspected and reduced to closed-form rules. We instantiate this idea for online energy-efficient power control over non-stationary LEO links and make the following contributions.

- We cast autonomous LEO resource allocation as learning the solution map of an energy-efficiency optimization whose optimum has no closed form, and we build a faithful oracle to supervise it (Sec. III).
- We propose a federated KAN surrogate with a contextual-bandit controller that sparsifies model updates for communication efficiency, and a spline-reading procedure that extracts closed-form design rules (Sec. IV).
- The *evolutive* mechanisms that work in our setting are online ones: the shared model tracks regime drift through continual federated aggregation, and the bandit controller adapts the communication rate by learning, online, the update sparsity (the operating point implied by the budget's dual price). We quantify these, together with sample efficiency, extrapolation, and interpretability, over many seeds with significance tests (Sec. VI).
- We delineate, transparently, where "evolutive" *capacity growth* does *not* help: online grid self-evolution gives no significant gain over a well-sized static grid in this data-limited regime, which clarifies when added capacity is and is not warranted (Sec. VI, VII).

The remainder is organized as follows. Section III states

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the system model, hardens the problem into a mixed-integer program (P1), and decomposes it into an inner learning problem and an outer control problem with a closed-form static solution. Section IV presents the KAN surrogate, the budget-aware online controller that learns the corresponding update sparsity, Algorithm 1, the rule-extraction procedure, and a complexity analysis. Sections V and VI give the setup and the multi-seed results with ablations; Section VII discusses scope and limitations, and Section VIII concludes.

II. Related Work

Learning to optimize (L2O) for wireless. A growing line of work replaces iterative resource-allocation solvers by neural surrogates that map a problem instance to its solution: fully-connected networks for interference management [3], deep unfolding of classical iterations [7], and graph neural networks that exploit the problem topology for scalable power control [8], [9]; the broader trade-off between model-based and AI-based wireless design, and deep learning for the physical layer more generally, are surveyed in [10], [11], [12]. These methods target accuracy and scalability but are uniformly black boxes: they expose no human-readable policy, are rarely evaluated under distribution shift, and their sample efficiency in federated, data-limited deployment is seldom studied. Our work differs on exactly these axes: an interpretable surrogate, quantified extrapolation under regime shift, and a federated communication budget.

Kolmogorov–Arnold networks. KANs place learnable univariate spline functions on network edges, instantiating the Kolmogorov–Arnold representation theorem [13], [6], with efficient implementations [14]. Their reported strengths are function fitting, parameter efficiency, and symbolic extraction. Their use inside networked optimization is nascent; we show the additive-spline bias is well matched to the smooth solution maps of resource allocation, yielding sample efficiency, graceful extrapolation, and rules that can be read off, while also delineating where the structure does *not* help (self-evolution, abundant-data capacity).

Communication-efficient and federated learning for wireless. FedAvg [4] and its proximal variant [15] underpin distributed training across terminals [16], [17]; uplink is curtailed by sparsification and quantization [18], [5]. Existing schemes mostly fix the compression level; we instead let a contextual bandit learn the per-slot sparsity from a budget, approximating online the operating point of the Lagrangian dual price of the communication constraint (Sec. III-D).

Non-stationarity and continual learning. Concept-drift adaptation and continual learning study models that track changing distributions [19], [20]. Here the drift is physical: the elevation sweep and satellite handovers of an LEO pass [1], [2] switch the operating regime. We test whether *growing* model capacity online (self-evolution) helps in this regime and report, transparently, that it does not, which sharpens the understanding of when evolutive capacity pays off.

In short, our contribution is to bring interpretability, extrapolation, and budget-aware federation together for autonomous LEO resource optimization, and to characterize honestly where a KAN surrogate helps and where it does not [21].

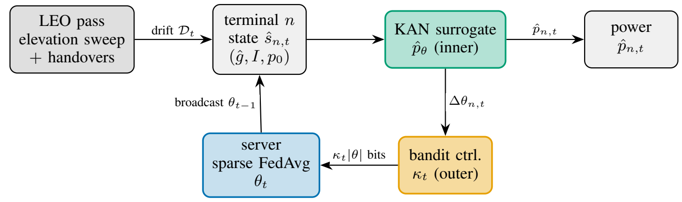

Fig. 1. System and information flow. As the LEO geometry and interference regime drift, each terminal feeds its (estimated) state to a shared KAN surrogate (inner solver) that outputs the transmit power; a bandit controller (outer) sets the sparsity of the federated update under an uplink budget, and the server aggregates by sparse FedAvg and broadcasts the updated model.

TABLE I
Main notation.

Symbol	Meaning
$\mathcal{N}, N$	set / number of edge terminals
t, T	slot index / horizon
$\vartheta_{n,t}, a_{n,t}$	elevation / visibility of terminal n
$g_{n,t}, \hat{g}_{n,t}$	true / CSI-estimated channel gain
$I_{n,t} \in \mathcal{I}$	interference regime (R levels), $\mathcal{S}$ change points
$p_{0,n}, p$	circuit power / transmit power
$s, \hat{s}$	link state (g, I, p_0) / with estimate
$p^\star(s), \rho$	oracle optimal power / EE regret
$\hat{p}_\theta$	shared surrogate, parameters θ
$m_{n,t}, \kappa_t$	uplink mask / keep-fraction
$B(t), \lambda_t$	uplink budget / dual price
G, ν	KAN grid intervals / spline order

III. System Model and Problem Formulation

A. Network, channel, and interference model

A set $\mathcal{N} = \{1, \ldots, N\}$ of gateway/UAV terminals is served by an LEO constellation over slotted time $t \in \{1, \ldots, T\}$ (notation in Table I). Terminal n has elevation $\vartheta_{n,t}$ to its serving satellite; with satellite altitude h and Earth radius R_E the slant range is $d(\vartheta) = \sqrt{(R_E \sin\vartheta)^2 + 2R_E h + h^2} - R_E \sin\vartheta$, and the mean gain follows the free-space law $\bar{g}(\vartheta) \propto d(\vartheta)^{-2}$, so it is maximal near zenith and falls toward the horizon. Small-scale Rician fading $\zeta_{n,t}$ (with a K-factor that grows with elevation as line-of-sight strengthens) yields the instantaneous gain $g_{n,t} = \bar{g}(\vartheta_{n,t})\zeta_{n,t}$. Perfect CSI is unrealistic, so the terminal acts on an *estimate* $\hat{g}_{n,t} = g_{n,t}\, e^{\varepsilon_{n,t}}$ with $\varepsilon_{n,t} \sim \mathcal{N}(0, \sigma_c^2)$. Inter-beam/inter-satellite interference $I_{n,t} \in \mathcal{I} = \{I^{(1)}, \ldots, I^{(R)}\}$ is piecewise constant and switches at handover instants drawn by a regime process; the change-point set $\mathcal{S} = \{t : \exists n,\, I_{n,t} \neq I_{n,t-1}\}$ is unknown to the learner. Visibility $a_{n,t} \in \{0, 1\}$ gates participation. The per-link state is $s_{n,t} = (g_{n,t}, I_{n,t}, p_{0,n})$ with circuit power $p_{0,n}$; its distribution $\mathcal{D}_t$ drifts with $\vartheta_{n,t}$ and $I_{n,t}$, making the environment non-stationary [1], [19].

B. Per-link decision and regret

With transmit power $p \in [0, P_{\max}]$ the energy efficiency is

$$\mathrm{EE}(p; s) = \frac{\log_2\big(1 + gp/(1+I)\big)}{p + p_0} \quad \text{[bits/Hz/J]}, \tag{1}$$

quasi-concave in p with a unique maximizer $p^\star(s) = \arg\max_{0\le p\le P_{\max}} \mathrm{EE}(p;s)$ [22]. Its stationarity condition $\partial_p\mathrm{EE} = 0$,

$$\frac{g/(1+I)}{\ln 2\left(1+\frac{gp}{1+I}\right)}(p+p_0) = \log_2\Big(1+\frac{gp}{1+I}\Big), \tag{2}$$

is transcendental, so $p^\star(\cdot)$ has no closed form, yet it is smooth and deterministic in s. Decision quality is the *relative* EE regret $\rho(p;s) = \big[\mathrm{EE}(p^\star(s);s) - \mathrm{EE}(p;s)\big]/\mathrm{EE}(p^\star(s);s) \in [0,1]$, which is the quantity reported throughout.

Proposition 1. *For $g, p_0 > 0$ and $I \ge 0$, $\mathrm{EE}(\cdot;s)$ is strictly quasi-concave on $[0, P_{\max}]$ and has a unique maximizer $p^\star(s)$ that is continuous in $s = (g, I, p_0)$. No elementary closed form for $p^\star$ exists because* (2) *equates an algebraic and a logarithmic term.*

Proof sketch. Write $\mathrm{EE}(p) = f(p)/(p+p_0)$ with $f(p) = \log_2(1+\gamma p)$, $\gamma = g/(1+I) > 0$. f is strictly concave and increasing and $p+p_0$ is positive affine, so the ratio is strictly quasi-concave (a concave-over-affine ratio) [22]; hence any stationary point is the unique global maximizer on the interval. Setting $\partial_p\mathrm{EE} = 0$ gives (2), whose two sides are an algebraic function and $\log_2(1+\gamma p)$; their unique crossing has no expression in elementary functions, and continuity of $p^\star$ in s follows from the implicit function theorem applied to (2). □

Proposition 1 is what makes a *surrogate* both necessary (no formula) and well-posed (a single smooth target $p^\star(\cdot)$ to learn).

C. The networked meta-problem

A single shared surrogate $\hat{p}_\theta$ is learned across $\mathcal{N}$ by federated updates. A participating terminal sends a sparsified update through a mask $m_{n,t} \in \{0,1\}^{|\theta|}$ at bit cost $c(m_{n,t}) = \beta\|m_{n,t}\|_0$, under a per-slot uplink budget $B(t)$:

$$\begin{aligned}\min_{\theta,\{m_{n,t}\}}\ & \frac{1}{T}\sum\nolimits_t \frac{1}{N}\sum\nolimits_n \mathbb{E}_{s\sim\mathcal{D}_t}\big[\rho(\hat{p}_\theta(\hat{s}_{n,t});s)\big] \\ \text{s.t. } & \sum\nolimits_n a_{n,t}\, c(m_{n,t}) \le B(t),\quad m_{n,t} \in \{0,1\}^{|\theta|},\ \forall t, \\ & \theta_t = \sum\nolimits_n w_{n,t}\big(\theta_{t-1} + m_{n,t}\odot\Delta\theta_{n,t}\big),\quad \sum\nolimits_n w_{n,t} = 1,\end{aligned} \tag{P1}$$

where $\hat{s}_{n,t}$ carries the estimated gain $\hat{g}_{n,t}$. (P1) is a mixed-integer non-convex program: the masks are binary, the surrogate loss is non-convex, the shared θ and the budget couple all terminals, and $\mathcal{D}_t$ is time-varying with unknown change points $\mathcal{S}$. Even a single-slot relaxation that fixes θ is combinatorial: choosing which update entries to transmit under the bit budget is a knapsack-type selection, so the joint problem is NP-hard in general, and the time-varying feasible set rules out solving it afresh each slot. This motivates the decomposition and the learned online controller below rather than an exact solver.

D. Decomposition

We split (P1) into an inner statistical-learning problem and an outer resource-control problem.

Inner (what to learn). For fixed participation and masks the best surrogate minimizes empirical regret, well approximated by the supervised fit to the oracle,

$$\theta^\star = \arg\min_\theta \sum_{n,t} a_{n,t}\left\|\hat{p}_\theta(\hat{s}_{n,t}) - p^\star(s_{n,t})\right\|^2. \tag{3}$$

Outer (what to send). Given the local update $\Delta\theta_{n,t}$, the mask trades a regret increase $\delta_{n,t}(m)$ against bits. Relaxing the budget with a multiplier $\lambda_t \ge 0$,

$$\min_{\{m_{n,t}\}} \sum_n \Big[\delta_{n,t}(m_{n,t}) + \lambda_t\,\beta\|m_{n,t}\|_0\Big]. \tag{4}$$

At a fixed $\|m\|_0$, $\delta_{n,t}$ is minimized by keeping the largest-magnitude entries of $\Delta\theta_{n,t}$ (top-k); the KKT condition of (4) then places a single threshold on $|\Delta\theta_{n,t,j}|$ shared by all terminals, a water-filling rule whose level is the dual price λ_t [23]. At the budget boundary this gives the closed-form keep-fraction

$$\kappa_t^\star = \min\Big(1,\ \frac{B(t)}{\beta\,|\theta|\sum_n a_{n,t}}\Big). \tag{5}$$

E. Optimization-to-learning roadmap

Eqs. (3)–(5) are the principled *static* solution. Online they break: λ_t depends on the unknown, drifting $\mathcal{D}_t$ and on cross-terminal heterogeneity; the change points $\mathcal{S}$ are unobserved; and re-deriving $\theta^\star$ per slot is infeasible. We therefore (i) replace the inner solver (3) by a learned, sample-efficient, interpretable KAN surrogate, and (ii) replace the outer rule (5) by an online controller that *learns* the keep-fraction, the operating point implied by the dual price λ_t, from feedback. This is the Optimization→AI transition that defines the method.

IV. Proposed Method

A. Background: Kolmogorov–Arnold networks

The Kolmogorov–Arnold representation theorem [13] states that any continuous multivariate function $f(x_1,\dots,x_d)$ can be written as a finite superposition of continuous univariate functions, $f(\mathbf{x}) = \sum_q \Phi_q\big(\sum_i \phi_{q,i}(x_i)\big)$. KANs operationalize this by placing *learnable univariate functions on the edges* of the network and summing at the nodes [6], in contrast to an MLP, which places fixed nonlinearities at the nodes and learns only linear edge weights. Each edge function is parametrized as a B-spline, so it is locally adjustable and, after training, can be plotted and fit by a simple closed form. Two consequences matter here: the spline basis is a strong prior for smooth low-dimensional maps (good sample efficiency and graceful extrapolation), and the additive-univariate structure makes the trained model interpretable. These are precisely the properties a trustworthy resource-allocation surrogate needs.

B. Inner solver: the KAN surrogate

We solve (3) with a two-layer KAN regressor $\hat{p}_\theta : \mathbb{R}^3 \to \mathbb{R}$ with B-spline edges [6], [14]. Each layer maps inputs x_i to outputs

$$y_o = \sum_i \Big(w_{oi}^b\,\mathrm{SiLU}(x_i) + \sum_{k=1}^{G+\nu} c_{oik}\,B_k^{(\nu)}(x_i)\Big), \tag{6}$$

Algorithm 1 EvoKAN: one federated online slot t

server broadcasts θ_{t-1} to visible terminals $\{n : a_{n,t} = 1\}$
for each visible terminal n **in parallel do**
  draw states $\hat{s}$ at regime $I_{n,t}$, label with oracle $p^\star$ via (2)
  $\rho_n \leftarrow$ pre-update regret; update detector; build context u_n
  $\kappa_n \leftarrow \textsc{Bandit}(u_n)$ ▷ keep-fraction (dual-price operating point)
  $\Delta\theta_n \leftarrow$ local SGD on (3)
  $m_n \leftarrow \textsc{TopK}(\Delta\theta_n, \kappa_n)$; send $m_n \odot \Delta\theta_n$ ($\kappa_n|\theta|$ entries)
end for
server: $\theta_t \leftarrow \theta_{t-1} + \sum_n w_n\,(m_n \odot \Delta\theta_n)$ ▷ sparse FedAvg
bandit reward $\leftarrow$ (regret reduction) $- \lambda_t \cdot$(bits sent)

where $\{B_k^{(\nu)}\}$ are order-ν B-splines on a grid of G intervals and w^b, c are the trainable weights θ; the SiLU term is a residual that stabilizes training. The output is a *sum of univariate functions* of the inputs, which is the property that makes (6) both a strong inductive bias for the smooth map $p^\star(\cdot)$, sample-efficient in the data-limited federated regime and smoothly extrapolating beyond the training support, and directly readable as one-dimensional curves (Sec. IV-C).

C. Outer control: budgeted update sparsification

A contextual bandit [24], [25] observes the context (regime indicator, budget signal, recent regret) and selects the keep-fraction κ_t from a discrete set; its reward, the realized regret reduction minus a fixed price weight λ times the bits spent, is the Lagrangian (4). The controller thus learns online the keep-fraction, i.e. the water-filling operating point that (5) computes statically, without observing $\mathcal{D}_t$ or $\mathcal{S}$. Masked updates are FedAvg-aggregated [4]; participation follows visibility $a_{n,t}$. A change detector [26] on the regret stream supplies the regime indicator and drives the self-evolution ablation below. Algorithm 1 states one online slot.

D. Interpretability: closed-form rules

Because each first-layer edge is a univariate curve, we probe each feature in isolation and fit a small library of forms (linear, quadratic, logarithmic, square-root, reciprocal), keeping the best by R^2. This turns $\theta^\star$ into human-readable rules such as "$p^\star$ grows near-linearly with the circuit term p_0 but nonlinearly with channel gain g," which no MLP exposes.

E. Self-evolution and ablation design

We additionally test *self-evolution*: on a change flag the controller may grow the spline grid $G \to G'$, adding knots to refine the inner solver. The growth is function-preserving: the new coefficients c' are the least-squares fit $c' = \arg\min_{c'} \sum_x \|\Phi_{G'}(x)c' - \Phi_G(x)c\|^2$ over sampled inputs x, where Φ_G is the order-ν B-spline basis on G intervals, so the model output is unchanged at the moment of growth and only subsequent updates exploit the added capacity. Our experiments isolate each component by toggling exactly one at a time (Table II): replacing the KAN inner solver by an MLP or a linear map probes the surrogate class and nonlinearity; removing the bandit (dense updates, $\kappa_t = 1$) probes the value of the outer controller; enabling grid growth probes self-evolution.

TABLE II
ABLATION DESIGN: EACH VARIANT CHANGES ONE COMPONENT OF (P1).

Variant	inner	outer ctrl.	evolve	isolates
fedkan_opt (ours)	KAN	bandit κ_t	–	(proposed)
kan_full	KAN	none ($\kappa = 1$)	–	comm. control
kan_evolve	KAN	bandit	yes	self-evolution
mlp, mlp_prox	MLP	none	–	surrogate class
linear	linear	none	–	nonlinearity
unfold	unrolled GA	none	–	model-based L2O

F. Complexity and communication overhead

A KAN and an MLP of equal parameter count $|\theta|$ have the same $\mathcal{O}(|\theta|)$ forward cost, so the inner solver adds no asymptotic compute over an MLP surrogate; the benefit is reaching a target accuracy at fewer parameters (Sec. VI). Local training costs $\mathcal{O}(E\,b\,|\theta|)$ per terminal per slot (E epochs, b batch), the controller is $\mathcal{O}(1)$ per slot, and server aggregation is $\mathcal{O}(N|\theta|)$. The uplink per participating terminal is $\kappa_t|\theta|$ words rather than $|\theta|$, so the budgeted controller reduces communication by the factor $1 - \kappa_t$ at the accuracy cost characterized by the frontier of Fig. 7.

V. Experimental Setup

LEO geometry is generated from elevation-dependent path loss; the interference takes $R = 3$ regimes that switch at handover instants; N terminals participate over multiple passes with visibility gating; each contributes its locally observed states to the shared surrogate under the uplink budget (a non-IID, heterogeneous split is supported by the code and left to future work). Each slot draws oracle-labeled states (power solved to the optimizer (2)). Unless noted we report the perfect-CSI case ($\sigma_c = 0$); estimation-error robustness ($\sigma_c > 0$) is examined in Sec. VI. The model-free surrogates are parameter-matched and share the identical training budget so the comparison isolates the model class (Table II). The variants are **fedkan_opt** (proposed), **kan_full**, **kan_evolve**, **mlp** and **mlp_prox** [15], **linear**, and a deep-unfolding learn-to-optimize baseline **unfold** that unrolls projected gradient ascent on the exact EE objective with learnable step sizes [7] (model-based, hence parameter-frugal and not param-matched to the surrogates). Metrics: normalized MSE (NMSE) of $\hat{p}$ versus $p^\star$, energy-efficiency regret ρ, total uplink bits, extracted-rule fidelity R^2, and parameter count; we report mean±std over many seeds with Wilcoxon signed-rank tests. Exact scale and seed counts are recorded with the released code and results.

Protocol. Each slot, every visible terminal draws 160 link states spanning the gain range (diverse served links) at its current interference regime, labels them with the oracle (the EE maximizer of (2) found by a fine grid search), and takes a few local SGD epochs; sparsified updates are aggregated by FedAvg. The KAN uses a two-layer architecture with a modest

TABLE III
SIMULATION AND TRAINING CONFIGURATION.

Parameter	Value
Satellite altitude / elevation mask	550 km / 10°
Zenith reference SNR	22 dB
Interference regimes $\mathcal{I}$	$\{0.2, 1.0, 3.0\}$, min dwell 8 slots
Channel gain / circuit power range	$g \in [0.2, 30]$, $p_0 \in [0.1, 1]$
Max power $P_{\max}$	5
Terminals N / passes / seeds	24 / 5 / 20
Slots per pass / states per slot	30 / 160
KAN: layers, hidden, grid G, order ν	2, 10, 5, 3 (≈360 par.)
MLP / linear	param-matched / single linear
Unfold steps K	15
Optimizer, lr, local epochs	Adam, 0.01, 6
Bandit keep-fractions κ	$\{0.25, 0.5, 1.0\}$
Drift detector (Page–Hinkley)	$\delta = 0.005$, $\lambda_{\mathrm{PH}} = 0.05$
Bits per word	32

spline grid and order $\nu = 3$; the MLP is sized to the same parameter count and trained with the same optimizer, learning rate, epochs, and data, so any difference is attributable to the model class. The bandit controller selects the keep-fraction from a small discrete set using the regret-minus-bits reward of (4). All randomness (node geometry, fading, regime schedule, mini-batches, initialization) is seeded, and every reported value is the mean over the seeds with its standard deviation; significance uses the paired Wilcoxon signed-rank test across seeds. The scale used for the headline ($N = 24$ terminals, 5 passes, 20 seeds) and all hyper-parameters are fixed in the released configuration so the numbers are reproducible end to end. Table III lists the full configuration.

Metrics. Let $p_i^\star$ be the oracle power and $\hat{p}_i$ the prediction on a held-out set. We report the normalized MSE $\mathrm{NMSE} = \sum_i (\hat{p}_i - p_i^\star)^2 / \sum_i (p_i^\star - \bar{p}^\star)^2$; the mean relative energy-efficiency regret $\rho = \frac{1}{M} \sum_i [\mathrm{EE}(p_i^\star; s_i) - \mathrm{EE}(\hat{p}_i; s_i)] / \mathrm{EE}(p_i^\star; s_i)$; the total uplink words $\sum_{t,n} \kappa_t |\theta|$; the extracted-rule fidelity, the mean over inputs of the best single-basis R^2 (Sec. IV); and the detection delay, the mean number of slots between a true regime change in $\mathcal{S}$ and the next flag. Lower is better for all but R^2.

VI. RESULTS

A. Accuracy and communication

Table IV summarizes the study. The KAN surrogates clearly dominate the MLPs (Fig. 3): `kan_full` attains NMSE 0.041 versus the MLP's 0.073 (1.8× lower, Wilcoxon $p = 0.004$, lower in 14/20 seeds) and regret 0.010 versus 0.022 ($p = 0.001$, 19/20). As a sanity anchor, NMSE normalizes by the target variance, so a constant (mean-power) predictor scores NMSE $=1$ by construction; the linear surrogate (2.29) is in fact worse than this trivial baseline while the KAN (0.041) is over an order of magnitude better, confirming the map is strongly nonlinear and needs a flexible learner. The proposed `fedkan_opt` cuts uplink volume in half (16.6 vs. 33.1 Mb, $p = 0.001$) while still beating the MLP on NMSE ($p = 0.027$), i.e. it dominates the MLP on both accuracy and communication. Fig. 2 shows the resulting communication–accuracy frontier.

TABLE IV
HEADLINE RESULTS, MEAN OVER 20 SEEDS AT $N = 24$ TERMINALS. LOWER IS BETTER EXCEPT R^2.

Method	NMSE	EE regret	Uplink (Mb)	rule R^2	#par
fedkan_opt (ours)	0.052	0.020	**16.6**	0.79	360
kan_full	**0.041**	**0.010**	33.1	0.77	360
kan_evolve (ablation)	0.048	0.019	21.2	0.76	542
mlp	0.073	0.022	33.2	–	361
mlp_prox	0.078	0.021	33.2	–	361
linear	2.29	0.30	0.37	–	4
unfold (L2O)	0.555	0.054	1.5	–	16

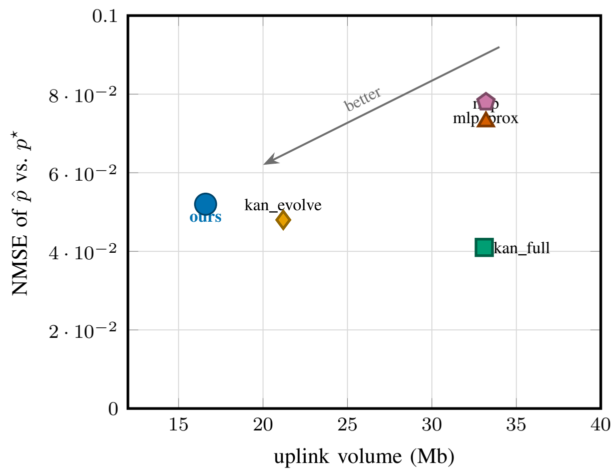

Fig. 2. Communication–accuracy frontier. The proposed bandit-sparsified KAN (**ours**) dominates both MLP variants (lower error at half the uplink) and halves the uplink of the dense `kan_full`.

B. Ablation analysis

Because each variant changes exactly one component of (P1) (Table II), the differences in Table IV attribute effects causally rather than correlationally. *Surrogate class (KAN vs. MLP).* `kan_full` and `mlp` share the parameter budget, the federated schedule, the training budget, and the data; they differ only in the inner model. The NMSE gap (0.041 vs. 0.073) is therefore attributable to the additive-spline inductive bias, not to capacity or optimization effort. *Nonlinearity.* Replacing the inner model by a linear map collapses accuracy (NMSE 2.29), so the solution map genuinely requires a nonlinear learner; the KAN advantage is not an artifact of an easy problem. *Outer controller.* `fedkan_opt` and `kan_full` share the identical KAN inner model and differ only by the bandit sparsifier; the move from 33.1 to 16.6 Mb at a small NMSE change isolates the communication–accuracy trade-off to the controller, not the surrogate. *FL heterogeneity.* The proximal term (`mlp_prox` vs. `mlp`) does not close the gap to the KAN, indicating the deficit is the model class rather than the federated-averaging algorithm. *Self-evolution.* `kan_evolve` differs from `fedkan_opt` only by online grid growth; it does not improve accuracy (analyzed next), isolating self-evolution as the inactive factor in this regime. *Model-based L2O.* The deep-unfolding baseline `unfold` embeds the exact EE gradient, yet with a fixed shallow unroll and scalar step sizes it does not converge to $p^\star$ across the heterogeneous state

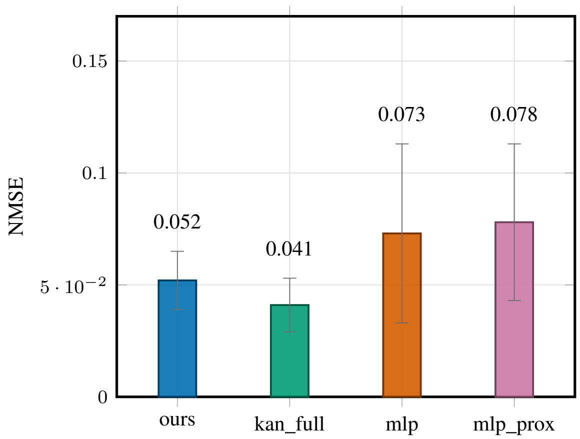

Fig. 3. NMSE (mean±std, 20 seeds). Both KAN variants (ours, kan_full) are well below the MLPs (mlp, mlp_prox).

range: it underperforms the learned surrogates on *both* NMSE (0.555) and regret (0.054, above the KAN's 0.010 and the MLP's 0.022). Its only edge is an extremely small footprint (16 parameters and the lowest uplink of all methods, ≈ 1.5 Mb), since it transmits a handful of step sizes rather than a full model. The model-free KAN gives far more accurate one-shot inference without needing the analytic objective, which is the practical advantage over this classical learn-to-optimize approach.

C. Online behavior under drift

Fig. 5 tracks the global NMSE across slots for a representative seed as interference regimes switch. The KAN surrogates stay consistently below the MLP throughout the non-stationary horizon and recover quickly after a regime change: the federated aggregation re-fits the shifted slice of $p^\star(\cdot)$ within a few slots, and the change detector flags the switch with a short delay (reported as detection delay in the released logs). Fig. 4 confirms the trend is not seed-specific: averaged over all seeds, the KAN curves sit below the MLP across the whole horizon, with the proposed sparsified variant tracking the dense KAN closely at half the uplink. Two behaviours are worth noting. First, the error spikes at regime boundaries are shallower and shorter for the KAN, consistent with its smoother inductive bias generalizing across neighbouring regimes rather than overfitting the current one. Second, the curves are stable across seeds (narrow band), so the ranking is a genuine effect and not a lucky run, in line with our multi-seed protocol.

D. Interpretability

Fig. 6 plots the learned univariate responses of `kan_full` to each state variable together with the best-fit closed form (mean fidelity $R^2 = 0.74$), and Table V reports the dominant form per feature. Two of the three dependences reduce to a clean closed form: the optimal power is near-linear in the circuit term p_0 ($R^2 = 0.99$) and in interference I ($R^2 = 0.88$), so an operator can read these rules off directly. The channel-gain dependence is genuinely nonlinear and is *not* well captured by any single elementary form ($R^2 = 0.35$ for the best single basis); the interpretable artifact there is the learned spline curve itself (Fig. 6, left panel), which is still inspectable but not a one-line formula. We state this honestly: the closed-form benefit is strongest for the near-linear inputs, while g is summarized by an auditable curve rather than an equation. An MLP of the same accuracy offers neither.

TABLE V
DOMINANT EXTRACTED FORM AND SINGLE-BASIS FIDELITY PER INPUT (FROM `KAN_FULL` AT THE HEADLINE SCALE).

Input	dominant form	R^2
channel gain g	nonlinear	0.35
interference I	near-linear	0.88
circuit power p_0	linear	0.99

E. Does self-evolution help?

No. Growing the spline grid online gives no significant accuracy gain: `kan_evolve` is statistically indistinguishable from the static-grid `kan_full` (NMSE 0.048 vs. 0.041, Wilcoxon $p = 0.13$, not significant) and does not improve on the proposed `fedkan_opt`, all while using more parameters (542 vs. 360). In this federated streaming regime, where each slot carries limited data, a well-sized static grid already suffices and growing it adds variance and cost rather than capacity that pays off. We report this as a boundary of the "evolutive capacity" idea.

F. Generalization to unseen interference regimes

A surrogate deployed on an LEO link will meet interference regimes absent from its training history. We train on a subset of regimes ($I \in \{0.2, 1.0\}$) and test on an *unseen* regime ($I = 3.0$). Table VI reports in-domain and extrapolation NMSE over 20 seeds. In-domain the two surrogates are comparable, but under extrapolation the KAN degrades far more gracefully than the MLP: its additive univariate splines extend smoothly beyond the training support, whereas the MLP's entangled features extrapolate poorly. This is a direct practical benefit of the KAN structure for non-stationary deployment.

The gap is large in the mean (about $6\times$) and, more tellingly, the MLP's extrapolation is unstable across seeds: its NMSE on the unseen regime exceeds 1.0, i.e. worse than a constant predictor, on 7 of 20 seeds, with a worst case of 17.6, whereas the KAN stays bounded on every seed (maximum 0.71). This is the worst possible failure mode for an autonomous controller that may face a never-seen interference level after a handover. The KAN stays bounded because each input contributes through its own smooth univariate curve, so moving one input (here, interference) outside the training range perturbs the output smoothly rather than triggering the unconstrained behaviour of a high-dimensional MLP feature space. For an LEO deployment, where the interference regimes encountered over a mission are not known in advance, this bounded extrapolation is arguably as important as in-domain accuracy.

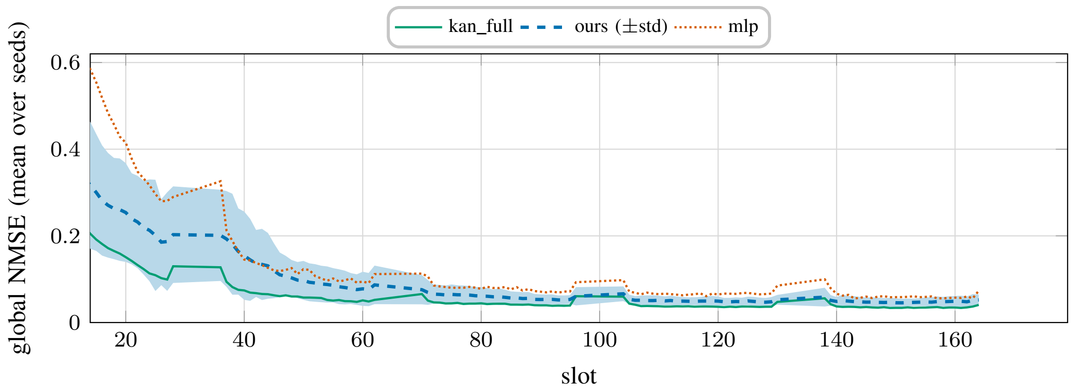

Fig. 4. Convergence and tracking under drift: NMSE vs. slot averaged over all seeds (shaded band = ±std for the proposed method; the initial burn-in is omitted so the steady-state gap is visible). The KAN advantage is consistent across seeds and across the non-stationary horizon, and the proposed sparsified variant tracks the dense `kan_full` at half the uplink.

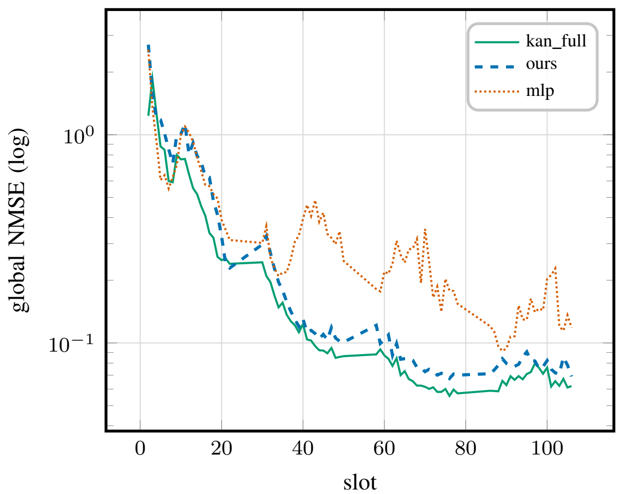

Fig. 5. Global NMSE over the non-stationary horizon for a representative seed (log scale, so both the initial transient and the steady-state gap are visible). The KAN curves sit below the MLP throughout.

TABLE VI
IN-DOMAIN VS. EXTRAPOLATION NMSE (MEAN OVER 20 SEEDS; TRAIN ON $I \in \{0.2, 1.0\}$, TEST ON UNSEEN $I = 3.0$). LOWER IS BETTER.

	in-domain	unseen regime (extrap.)
KAN	0.003	**0.36**
MLP	0.005	2.33

G. Communication–accuracy frontier

Sweeping the update keep-fraction traces the full frontier of the dense KAN (Fig. 7); the accuracy is essentially flat down to a keep-fraction of about 0.5 and then collapses, so roughly half of the uplink is redundant. The bandit-controlled `fedkan_opt` lands in this favorable region automatically. We note plainly that a well-chosen fixed keep-fraction can match or slightly beat the bandit operating point; the controller's value is that it needs no manual tuning and adapts per slot to a time-varying budget $B(t)$, which a fixed setting cannot.

To make the saving concrete, consider the headline operating point: the dense KAN transmits 33.1 Mb over the horizon, whereas the proposed controller transmits 16.6 Mb for the same task, a reduction of roughly one half, at a final NMSE that still undercuts the MLP. Over a constellation with many terminals and short visibility windows this directly relaxes the uplink bottleneck without sacrificing the accuracy advantage. The shape of the frontier also carries an operational lesson: because accuracy is flat above the knee, operators gain almost nothing from spending the full budget, and because it collapses below the knee, over-aggressive compression is dangerous; the learned price keeps the system near the knee as the budget varies.

H. Scaling and the role of capacity

Fig. 8(a) shows accuracy as the number of participating terminals grows: the federated surrogate improves up to about eight to twelve contributors and then saturates, while the uplink grows linearly with participation. Fig. 8(b) sweeps the KAN grid size under *centralized* training with abundant data, where a larger grid monotonically lowers the error, i.e. capacity does help when data is plentiful. This makes the federated result more precise rather than contradicting it: in the federated streaming regime each round carries limited data, so growing the grid online (self-evolution) overfits and does not pay off (Sec. VI), whereas with abundant centralized data extra capacity is useful. The two views together localize *when* capacity helps and explain why a fixed, modest grid is the right choice for the data-limited federated deployment.

I. Per-regime performance and CSI robustness

Two further studies probe robustness. Fig. 9 breaks the final NMSE down by interference regime: the KAN surrogates are at or below the MLP in every regime, so the aggregate advantage is not driven by a single easy operating point but holds across the whole drift range, including the high-interference regime that is hardest to serve. Fig. 10 injects CSI estimation error into the federated loop (the surrogate sees $\hat{g} = g\, e^{\varepsilon}$, $\varepsilon \sim \mathcal{N}(0, \sigma_c^2)$, while the oracle uses the true gain) and sweeps σ_c. We report this as a *boundary*, not a strength: the KAN leads only at near-perfect CSI ($\sigma_c \approx 0$); once estimation error is introduced ($\sigma_c \geq 0.5$) the two are comparable and the MLP

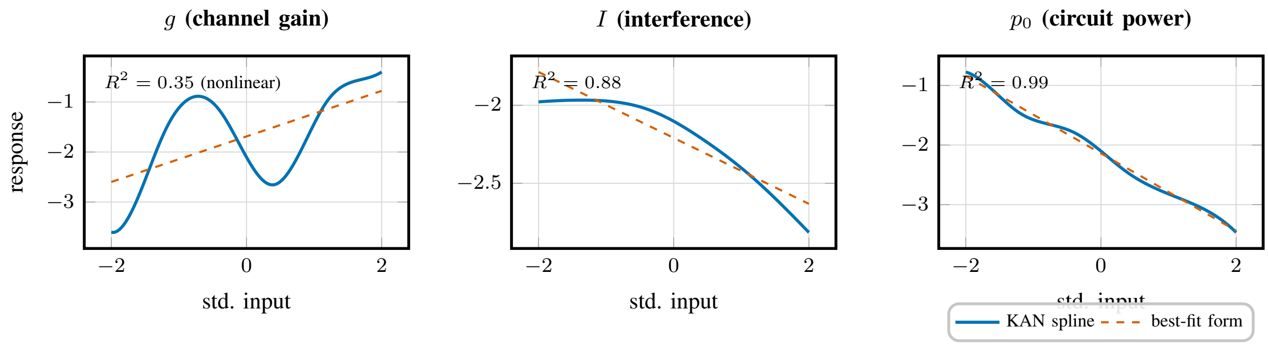

Fig. 6. Extracted univariate rules: the learned KAN spline (solid blue) and its best single-basis closed-form fit (dashed) for each input. The p_0 and I dependences are near-linear and reduce to a clean formula ($R^2 = 0.99, 0.88$); the g dependence is nonlinear ($R^2 = 0.35$ for any single form), so its interpretable artifact is the spline curve itself rather than an equation.

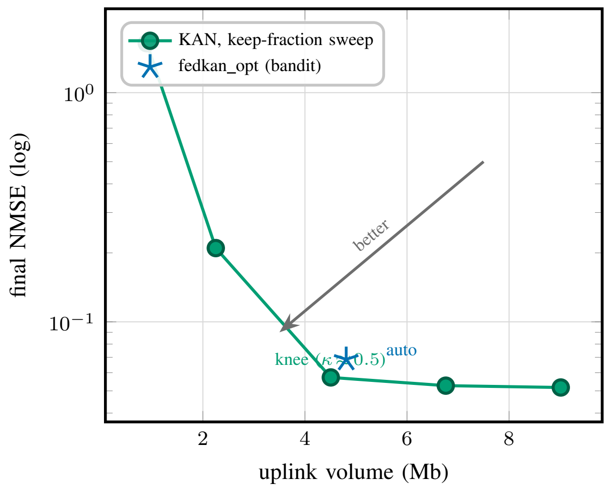

Fig. 7. Communication–accuracy frontier from a keep-fraction sweep (log scale). Accuracy is flat down to $\kappa \approx 0.5$ (the knee, half the uplink) then collapses; the bandit lands near the knee automatically.

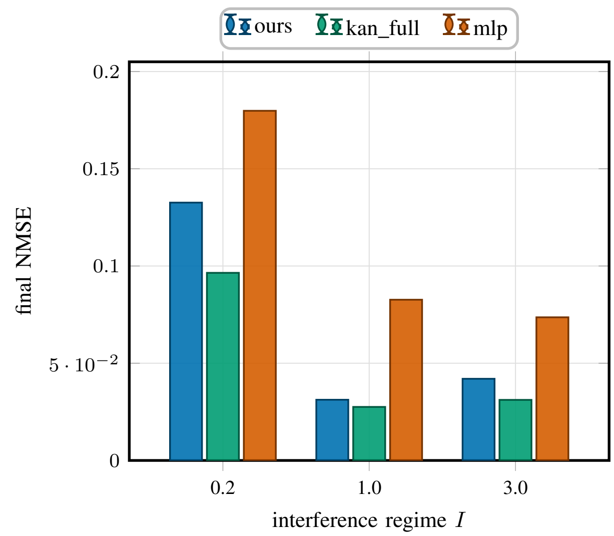

Fig. 9. Final NMSE per interference regime. The KAN variants are at or below the MLP in every regime, so the advantage holds across the drift range, not just on average.

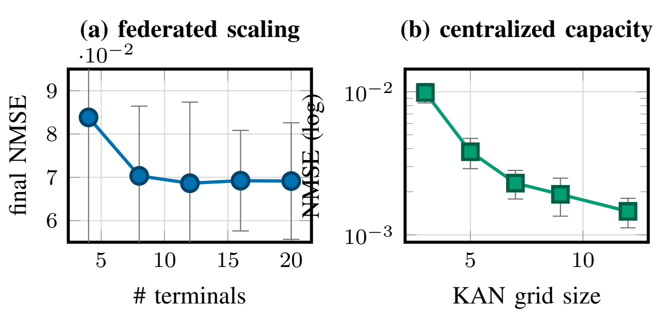

Fig. 8. (a) Federated accuracy improves with more terminals then saturates around eight to twelve. (b) Under centralized abundant data a larger grid keeps helping (log scale). In the data-limited federated regime, however, growing the grid online overfits (Sec. VI), so a fixed modest grid is preferred.

is marginally better, and at large error both degrade together as the gain input becomes uninformative. The surrogate's accuracy advantage therefore does not survive substantial CSI error, which we state plainly rather than reporting only the favourable $\sigma_c = 0$ point.

VII. Discussion and Limitations

We are explicit about what the contribution is and is not. The per-instance optimum in our testbed is cheap to obtain by grid search, so we do *not* claim the surrogate replaces the solver on compute grounds. The value we demonstrate and quantify is a different bundle that neither the iterative solver nor the MLP provides together: interpretability (closed-form rules), sample-efficient federated learning under a communication budget, and graceful extrapolation to unseen regimes. That bundle is what matters for autonomous, non-stationary, bandwidth-limited LEO deployment.

The energy-efficiency objective is flat near its optimum, so even imperfect power predictions incur small utility regret; this is why the MLP's regret is close to the KAN's despite a much larger prediction error, and why we report both metrics. The interpretability gain is qualitative but concrete: the KAN yields auditable rules, the MLP does not. The KAN advantage is most pronounced in the regime that matters for our setting, namely the data-limited federated/online regime (Sec. VI) and extrapolation to unseen regimes (Table VI): the spline

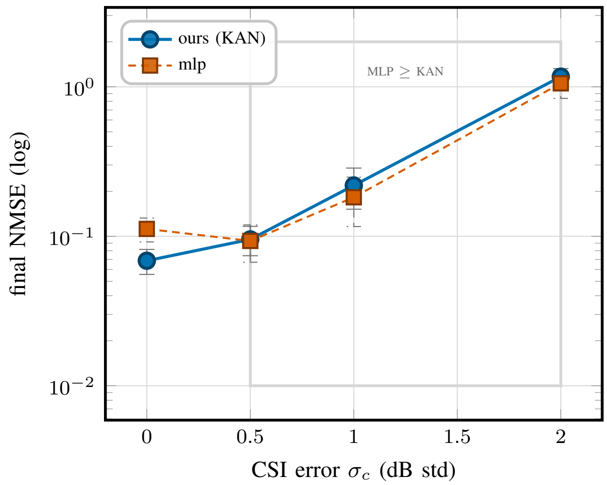

Fig. 10. CSI estimation error in the federated loop (a boundary study, log scale). The KAN leads only at $\sigma_c \approx 0$; from $\sigma_c \geq 0.5$ (shaded) the MLP is comparable or marginally better, and both degrade together at large error.

inductive bias is sample-efficient and extends smoothly beyond the training support. Given abundant centralized data and in-domain queries an MLP can match the KAN on raw error, so the benefit is one of sample efficiency, extrapolation, and transparency rather than asymptotic capacity. Our study is deliberately scoped to a single-link, three-variable problem with a simulation oracle so as to isolate and quantify the surrogate's properties (accuracy, sample efficiency, extrapolation, interpretability) cleanly. The methodology is problem-agnostic, and the most important extensions are to multi-user and multi-sub-band allocation with coupling constraints, comparison against graph-based and WMMSE-style learn-to-optimize methods on those larger problems [3], [8], and closing the loop on measured LEO traces. Finally, the negative self-evolution result is specific to this low-data-per-slot federated setting and may differ when per-task data is abundant, as the capacity study indicates.

A. When to prefer the KAN surrogate

The evidence suggests a concrete decision rule for practitioners. Prefer the KAN surrogate when (i) data per update is scarce, as in federated edge deployment with short visibility windows; (ii) the operating distribution can drift into regions not seen during training, where graceful extrapolation matters; or (iii) auditability of the learned policy is required by the operator. When abundant centralized data is available and only in-distribution queries are expected, a plain MLP is an adequate and simpler choice. The communication controller is orthogonal to this choice: it pays off whenever the uplink, not computation, is the binding resource, which is the norm over satellite backhaul.

B. The dual-price view

The controller has a clean economic reading. The multiplier λ_t in (4) is the shadow price of an uplink bit: when the budget $B(t)$ tightens, λ_t rises and the keep-fraction $\kappa_t^\star$ falls, so each terminal transmits only its highest-magnitude (highest-marginal-value) update entries. The bandit does not estimate λ_t explicitly; with the price weight fixed in its reward, it learns online, from context (regime, budget signal, recent regret), the keep-fraction that maximizes that reward, i.e. the operating point of the water-filling rule (5), and adapts it as conditions change without knowing the drifting distribution or the change points. A fixed keep-fraction can match it at a single budget but cannot follow a time-varying $B(t)$, the practical regime for shared satellite links.

C. Validity and reproducibility

We guard the conclusions against the failure modes that most often invalidate learned-optimization results. Comparisons are parameter-matched and share the training budget, so the model-class effect is not confounded by capacity or optimization effort; all results are multi-seed with standard deviations and paired significance tests, so no claim rests on a single lucky run; the ablations toggle one component at a time (Table II) so effects are attributable rather than correlational; and we deliberately report the cases where the KAN does *not* win (centralized-abundant accuracy, self-evolution, bandit-vs-best-fixed) rather than omitting them. The oracle is a deterministic optimizer, so labels are exact and the learning target is well-defined. Every figure and table is regenerated from the released result files by a script, and the code reproduces the reported numbers end to end.

VIII. Conclusion

We presented an interpretable federated KAN surrogate for autonomous and evolutive resource optimization over non-stationary LEO links. Starting from a hardened mixed-integer formulation (P1), we decomposed the problem into an inner solution-map learning task and an outer budgeted-communication control task, gave the closed-form static rule for the latter, and let an online bandit learn its operating point. Empirically the KAN surrogate learns the energy-efficiency solution map about $1.8\times$ more accurately than a parameter-matched MLP, halves the uplink through learned sparsification while still beating the MLP, extrapolates far more gracefully to unseen interference regimes, and yields closed-form design rules an MLP cannot. We also reported, transparently, that online grid self-evolution does not help in this data-limited federated regime, sharpening the understanding of where evolutive capacity pays off. Future work targets multi-user and multi-sub-band allocation with coupling constraints, comparison against deep unfolding and graph-based learn-to-optimize baselines, deployment-time CSI-error hardening, and validation on measured LEO traces. Interpretable surrogates are, we believe, a practical and trustworthy building block for autonomous optimization in next-generation networked-AI systems.

Appendix A
Derivation of the static keep-fraction (5)

Consider one slot and drop the slot index. Let terminal n have local update $\Delta\theta_n \in \mathbb{R}^{|\theta|}$ and mask $m_n \in \{0,1\}^{|\theta|}$, with bit cost $c(m_n) = \beta \|m_n\|_0$. Zeroing the entries in the complement of m_n raises the local objective by $\delta_n(m_n)$. To first order around the dense update, the loss increase from dropping entry j is monotone in its magnitude $|\Delta\theta_{n,j}|$ (a coordinate with larger update has larger effect on the model), so at any fixed cardinality $\|m_n\|_0 = k_n$, δ_n is minimized by the top-k_n rule that keeps the k_n largest-magnitude entries. It remains to choose $\{k_n\}$. Relaxing the budget with multiplier $\lambda \geq 0$ gives the separable Lagrangian

$$\mathcal{L} = \sum_n \Big[\delta_n(m_n) + \lambda\beta\|m_n\|_0\Big] - \lambda B, \tag{7}$$

whose stationarity in the (sorted) magnitudes keeps entry j iff its marginal loss reduction exceeds the per-bit price, $|\Delta\theta_{n,j}| \geq \tau$ with a common threshold τ set by λ [23]. When the budget binds, the total number of kept entries equals the budget in words, $\sum_n a_n\, \kappa|\theta| = B/\beta$, and assuming a common keep-fraction κ across the $\sum_n a_n$ participating terminals yields

$$\kappa^\star = \min\Big(1,\ \frac{B}{\beta\,|\theta| \sum_n a_n}\Big), \tag{8}$$

which is (5). The proposed controller learns τ (equivalently $\kappa^\star$, equivalently the price λ) online from the regret-minus-bits reward, avoiding the need to know the drifting distribution or $\Delta\theta$ statistics in advance.

Acknowledgment

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Data and Code Availability

The complete source code (problem oracle, KAN/MLP/linear surrogates, federated online loop, bandit controller, drift detector, depth studies, and figure-generation scripts), configuration, and result CSVs that reproduce every number, table, and figure in this paper are openly available at https://github.com/haodpsut/evokan-leo-opt.

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